



```
In [4]: !pip install groq
```

```
Collecting groq
  Downloading groq-1.5.0-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from groq) (4.14.0)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from groq) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from groq) (0.28.1)
Requirement already satisfied: pydantic<3,>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from groq) (2.13.4)
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from groq) (1.3.1)
Requirement already satisfied: typing-extensions<5,>=4.14 in /usr/local/lib/python3.12/dist-packages (from groq) (4.15.0)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5,>=3.5.0->groq) (3.18)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->groq) (2026.6.17)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->groq) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->groq) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->groq) (0.7.0)
Requirement already satisfied: pydantic-core==2.46.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->groq) (2.46.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->groq) (0.4.2)
Downloading groq-1.5.0-py3-none-any.whl (143 kB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0.0/143.7 kB ? eta -:-:-:-
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 143.7/143.7 kB 4.3 MB/s eta 0:00:00

Installing collected packages: groq
Successfully installed groq-1.5.0
```

```
In [19]: from groq import Groq
        from google.colab import userdata
```

```
In [20]: api_key = userdata.get("GROQ_API_KEY")

        client = Groq(api_key=api_key)
```

```
In [29]: # Initial Prompt
        prompt = "Explain Generative AI in 50 words"

        response = client.chat.completions.create(
            model="llama-3.3-70b-versatile",
            messages=[
                {
                    "role": "user",
                    "content": prompt
                }
            ]
        )
```

```

],
temperature=0.5,
max_tokens=100
)
generated_text1 = response.choices[0].message.content
generated_text1

```

Out[29]: 'Generative AI creates original content, such as images, music, and text, using algorithms and machine learning models, allowing it to generate new, unique, and often realistic outputs, mimicking human creativity and imagination.'

```

In [30]: # Modified prompt
new_prompt = "Explain Machine Learning in 50 words"

response2 = client.chat.completions.create(
    model="llama-3.3-70b-versatile",
    messages=[
        {
            "role": "user",
            "content": new_prompt
        }
    ],
    temperature=0.5,
    max_tokens=100
)

generated_text2=response2.choices[0].message.content
generated_text2

```

Out[30]: 'Machine Learning is a subset of AI that enables systems to learn from data, identify patterns, and make predictions or decisions without explicit programming, improving performance over time through experience and feedback.'

Working of the Groq LLM API

Input

The user provides a prompt such as "Explain Generative AI in 50 words". This prompt is sent to the Groq API along with the selected language model.

Processing

The Groq API receives the request and forwards it to the specified Large Language Model (LLM). The model analyzes the prompt, understands its context, and generates an appropriate response using its trained knowledge.

Output

The API returns a structured response containing metadata and the generated text.
The generated text is extracted from the response object using:

```
response.choices[0].message.content
```

The extracted text is then displayed to the user.